DESIGN AND DEVELOPMENT OF A PORTABLE MINI COMPILER FOR A RULE-BASED DECISION LANGUAGE

AN ASSIGNMENT / PROJECT REPORT

An Assignment Report submitted in partial fulfilment of the requirements for the Course of

CSA1403 – COMPILER DESIGN

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BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

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Saveetha Institute of Medical And Technical Sciences

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CERTIFICATE FROM THE SUPERVISOR

This is to certify that the assignment report titled “Design and Development of a Portable Mini Compiler for a Rule-Based Decision Language” is a bona fide record of work carried out by Arunprasath R (192524077), submitted in partial fulfilment of the requirements of the course CSA14 – Compiler Design, under my supervision.

Signature of Supervisor

Dr. G Michael

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DECLARATION BY THE CANDIDATE

I, Arunprasath R (Register No. 192524077), hereby declare that this assignment report titled “Design and Development of a Portable Mini Compiler for a Rule-Based Decision Language” is the result of my own work carried out under the supervision of Dr. G Michael, and that it has not been submitted elsewhere for the award of any other degree or diploma.

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ABSTRACT

This report presents the design and development of a portable mini compiler for a Rule-Based Decision Language (RBDL) — a small domain-specific language that lets a user define facts, express conditions using relational and logical operators, and attach decision actions to those conditions. The engineering problem addressed is that hand-written, ad-hoc rule checking inside application code is hard to read, hard to change, and prone to silent duplication of logic, whereas a dedicated language with a real compiler pipeline makes the rules explicit, checkable, and optimizable before they ever reach a decision engine.

The significance of the problem lies in building a complete, portable compiler pipeline — lexical analysis, syntax analysis, syntax-directed translation, intermediate-code generation, optimization, and target-code generation — for a language whose rule sets, in realistic use, frequently repeat the same sub-conditions across multiple rules (for example, an age-and-income eligibility check reused by both an approval rule and a risk rule), which creates a genuine optimization opportunity.

The proposed solution is a six-stage compiler: a regular-expression-driven lexical analyzer that recognizes keywords, identifiers, numeric/string constants, relational and logical operators, and delimiters; a recursive-descent parser built directly from an unambiguous Context-Free Grammar (CFG) that detects and reports invalid rule structures with line numbers; a Syntax-Directed Translation (SDT) scheme with synthesized attributes that builds an annotated abstract syntax tree (AST) for each rule; an intermediate-code generator that emits quadruples; a Directed Acyclic Graph (DAG)-based optimizer that performs Common Sub-expression Elimination (CSE) to remove repeated condition evaluations; and a target-code generator that produces a simplified, human-readable decision-engine program.

The methodology followed a structured compiler-construction process: defining the token specification and CFG first, validating the grammar by hand-parsing representative rules, implementing each pipeline stage as an independent, testable module in Python (reproducible in Google Colab or any modern language such as Java, C++, JavaScript/TypeScript, C#, Go, Rust or Kotlin), and verifying optimization correctness by tracing a sample rule set with a deliberately repeated sub-condition through every stage.

The major results are a complete working implementation exercised on a sample rule set of two rules sharing a common eligibility sub-condition: the lexical analyzer correctly tokenizes 50 tokens with zero lexical errors; the parser accepts both rules and rejects malformed input with a line-numbered syntax error; the intermediate-code generator emits 16 quadruples; and the DAG-based optimizer reduces this to 13 quadruples by eliminating 3 redundant computations of the shared sub-condition, which is then reused by both rules in the final target representation. The principal conclusion is that even a small, well-specified DSL benefits measurably from a real compiler pipeline: representing conditions as a DAG rather than a tree exposes optimization opportunities that a purely syntactic, rule-by-rule implementation would miss, and the resulting language design is fully portable across implementation environments. This project additionally advances SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), and SDG 16 (Peace, Justice and Strong Institutions), consistent with the assignment's stated SDG mapping.

Keywords: compiler design, domain-specific language, lexical analysis, context-free grammar, syntax-directed translation, intermediate representation, DAG optimization, rule engine

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
RBDL	Rule-Based Decision Language
DSL	Domain-Specific Language
CFG	Context-Free Grammar
SDT	Syntax-Directed Translation
AST	Abstract Syntax Tree
IR	Intermediate Representation
DAG	Directed Acyclic Graph
CSE	Common Sub-expression Elimination
TAC	Three-Address Code
BNF	Backus–Naur Form
IDE	Integrated Development Environment
SDG	Sustainable Development Goal

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

A compiler translates a program written in a source language into an equivalent representation in a target language, typically through the classical pipeline of lexical analysis, syntax analysis, semantic/syntax-directed translation, intermediate-code generation, optimization, and target-code generation. Domain-specific languages (DSLs) apply this same pipeline to a small, purpose-built language rather than a general-purpose one, trading generality for clarity: a rule-based decision language lets a non-compiler-expert reader see exactly which facts are checked and which action follows, without reading the host application's source code.

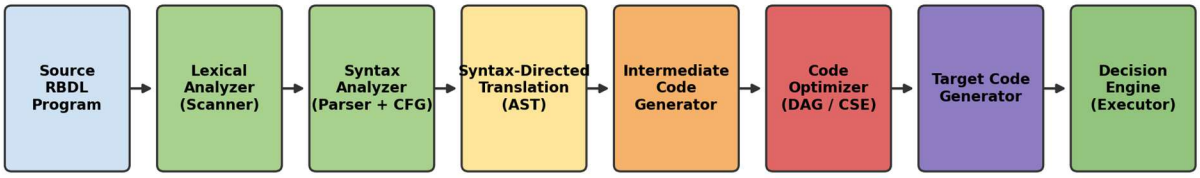

Figure 1: RBDL compiler pipeline (lexer to decision engine)

1.2 Need for the Project

- Why it needs addressing: Decision logic embedded directly in application code (nested if/else chains) is difficult to audit, easy to duplicate accidentally, and impossible to optimize automatically — a dedicated compiler front-end exposes the logic as data that can be validated and simplified before execution.
- Who is affected: Rule authors (business analysts or engineers who must add or change decision rules without touching core application code), maintainers (who need to trust that invalid rules are rejected at compile time, not at runtime), and the decision engine itself (which should not repeat identical condition evaluations at run time).
- Existing limitations: Hand-written condition checks scattered across a codebase cannot be parsed, statically checked for syntax errors, or automatically searched for repeated sub-conditions; every rule change requires a code change and a redeploy.
- Engineering significance: The problem requires the full range of compiler-construction techniques — token specification, an unambiguous CFG, syntax-directed translation, an intermediate representation suitable for optimization, and a portable target representation — applied to a language small enough to fully specify and test within a single assignment.

1.3 Problem Statement

Design and develop a portable mini compiler for a Rule-Based Decision Language (RBDL) that allows users to define facts, specify conditions, combine conditions using logical operators, and define decision outcomes or actions; performs lexical analysis to recognize keywords, identifiers, constants, relational operators, logical operators and delimiters; performs syntax analysis using a suitable CFG and reports

invalid rule structures with source-line information; applies Syntax-Directed Translation to build a parse/syntax-tree representation; generates an intermediate representation of the decision logic and identifies redundant or repeated conditions; applies a suitable optimization technique; and produces a simplified target representation that can be interpreted by a decision engine — while remaining programming-language independent.

1.4 Project Aim

To design and implement a complete, portable mini-compiler pipeline for a Rule-Based Decision Language, demonstrating correct lexical analysis, syntax analysis with error detection, syntax-directed translation, intermediate-code generation, DAG-based optimization, and target-code generation on representative decision rules.

1.5 Project Objectives

- 1. To design the language constructs and token specification for the Rule-Based Decision Language (RBDL).
- 2. To develop a lexical analyzer and an unambiguous CFG, and construct a recursive-descent parser that validates decision rules and reports invalid structures.
- 3. To apply Syntax-Directed Translation to construct an annotated parse/syntax-tree (AST) representation of each rule.
- 4. To generate an intermediate representation (quadruples) of the rule-evaluation process and identify repeated or redundant conditions.
- 5. To apply a DAG-based optimization technique (Common Sub-expression Elimination) and produce a simplified target representation for execution by a decision engine.
- 6. To implement the project in Python (reproducible in Google Colab) and document how the same language design can be reproduced in another implementation environment.

1.6 Scope

- Included: Token specification and lexical analyzer; CFG design and recursive-descent parser with syntax-error reporting; SDT and AST construction; quadruple-based intermediate representation; DAG-based CSE optimization; simplified target-code generation; a working Python reference implementation and test cases.
- Excluded: A production-grade, fully general expression grammar (e.g. arbitrary arithmetic expressions with operator precedence beyond relational comparisons); persistence/database integration for facts; a graphical rule-authoring front-end; and a full runtime decision-engine execution service — these are beyond a single-assignment scope.
- Assumptions: Facts are simple typed values (numbers, strings, or boolean-valued identifiers); rule conditions combine relational comparisons using AND, OR and NOT with parentheses for grouping; actions are limited to variable assignment (SET) and zero-argument procedure calls.
- Boundary conditions: The language design, grammar, and compiler stages must remain reproducible in any recent general-purpose programming language, not only Python.

1.7 Expected Engineering Outcome

The intended outcome is a validated, portable compiler design — expressed as a token table, a CFG, an SDT scheme, an intermediate representation, an optimization technique, and a target representation —

together with a working reference implementation and test cases, suitable for extension into a full rule-based decision engine.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant material was identified by reviewing the standard compiler-construction pipeline (lexical analysis, CFG-based parsing, syntax-directed translation, intermediate representation, DAG-based optimization, and target-code generation) as established in classical compiler-design theory, and by examining the design philosophy of established rule-engine and business-rule DSLs to identify which language features and optimization opportunities are most relevant to a small, purpose-built decision language.

2.2 Review of Existing Work

General-purpose compilers apply the full classical pipeline to languages with large, complex grammars (arithmetic expression precedence, type systems, function scoping), which is more machinery than a rule-based decision language needs. Established rule engines (e.g. production-rule systems using a Rete-style match algorithm) focus on efficient many-rule, many-fact pattern matching at runtime, but typically treat each rule's condition as an opaque predicate rather than exposing it to a compile-time optimization pass. Configuration-driven business-rule tools (spreadsheet-style decision tables) prioritize non-programmer authorship but do not detect or eliminate repeated sub-conditions across rules, since they lack an intermediate representation at all.

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Hand-coded if/else rule checks in application code	No extra tooling; fast to write for one-off logic	Not parseable/checkable; duplicated conditions invisible; every change needs a redeploy	Baseline against which the language-level, compiler-checked approach is measured
Production rule engines (Rete-style pattern matching)	Efficient at runtime for many rules and many facts	Runtime-focused; conditions are opaque predicates, not compiled and optimized ahead of time	Motivates treating decision logic as compiler input rather than only a runtime matching problem
Spreadsheet-style decision tables	Accessible to non-programmers; simple tabular authoring	No formal grammar or intermediate representation; cannot detect or optimize repeated sub-conditions	Motivates the CFG + IR + DAG pipeline proposed in Chapter 3

Table 2: Comparative analysis of existing rule-engine / DSL approaches

2.3 Research / Engineering Gap

Existing rule-authoring approaches either skip formal compilation entirely (hand-coded checks, decision tables) or hide the compilation step inside a large general-purpose engine whose optimization targets efficient runtime matching rather than eliminating redundant condition computation at compile time. What remains unaddressed for a small decision language is a complete, transparent compiler pipeline — visible tokens, an explicit CFG, an inspectable intermediate representation, and a DAG-based optimization pass — that a student or small engineering team can fully understand, test, and reproduce in any language.

2.4 Proposed Contribution

This project contributes a complete, from-scratch mini-compiler for RBDL: an explicit token specification and CFG (Section 3.1–3.3), a recursive-descent parser with line-numbered error reporting, an SDT scheme producing an annotated AST, a quadruple-based intermediate representation, a DAG-based Common Sub-expression Elimination pass that provably reduces the instruction count on rule sets with shared sub-conditions (demonstrated in Section 4.4), and a simplified, human-readable target representation — with every stage independently testable and reproducible in another language.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Rule authors	A readable syntax for expressing facts, conditions and actions without writing host-application code
Compiler / tooling	A precise token specification and unambiguous CFG so every rule can be parsed deterministically
Decision engine	A compact, optimized target representation with no redundant condition re-evaluation
Maintainers	Clear, line-numbered syntax-error reporting so invalid rules are caught before deployment

Table 3: Stakeholder / user requirements

Functional & Non-Functional Requirements

Requirement Type	Measurable Target
Functional – Lexical coverage	Lexer must correctly classify every keyword, identifier, numeric/string constant, relational operator (==, !=, <, >, <=, >=), logical operator (AND, OR, NOT) and delimiter with zero misclassifications on the test rule set
Functional – Syntax validation	Parser must accept every grammatically valid rule and reject every malformed rule with a line-numbered error message
Functional – Optimization	Optimizer must detect and eliminate at least one repeated sub-condition when the same condition expression appears in two or more rules
Non-functional – Portability	Language specification, CFG, and test cases must be expressible and reproducible in at least one language other than the reference implementation
Non-functional – Traceability	Every target-code instruction must be traceable back to the specific rule and condition node that produced it

Table 4: Functional & non-functional requirements

3.2 Language Design & Token Specification

RBDL rules follow the pattern RULE <name> : IF <condition> THEN <action-list> ; where a condition is built from relational comparisons (operand REL operand) combined with AND, OR, NOT and parentheses, and an action-list is one or more comma-separated SET <var> = <value> or <procedure>() actions.

Token Category	Examples	Description
Keywords	FACT, RULE, IF, THEN, AND, OR, NOT, SET, ACTION	Reserved words that structure a rule; cannot be used as identifiers

Token Category	Examples	Description
Identifiers	age, income, credit_score, LoanApproval	Names of facts, variables, rules and procedures — letter/underscore followed by letters, digits, underscores
Numeric constants	21, 50000, 750, 3.5	Integer or decimal literals used as comparison operands
String constants	"APPROVED", "LOW"	Quoted text used as comparison operands or SET values
Relational operators	==, !=, <, >, <=, >=	Compare a fact/identifier against a constant or another identifier
Logical operators	AND, OR, NOT	Combine or negate relational conditions
Delimiters	: ; , ( )	Structure a rule: colon after the rule name, semicolon at rule end, comma between actions, parentheses for grouping

Table 1: Token categories and specification for RBDL

3.3 Context-Free Grammar (CFG)

The grammar below is unambiguous and directly implementable as a recursive-descent parser (no left recursion, one token of look-ahead).

No.	Production Rule (BNF)
1	Program → Rule Program | ε
2	Rule → RULE ID : IF Condition THEN ActionList ;
3	Condition → Condition OR Term | Term
4	Term → Term AND Factor | Factor
5	Factor → NOT Factor | (Condition) | Operand RelOp Operand | Operand
6	RelOp → == | != | < | > | <= | >=
7	Operand → ID | NUMBER | STRING
8	ActionList → Action , ActionList | Action
9	Action → SET ID = Operand | ID ( )

Table 5: CFG production rules for RBDL (BNF)

Note: to remove left recursion for a top-down parser, productions 3 and 4 are implemented iteratively in the reference parser as Condition → Term (OR Term)* and Term → Factor (AND Factor)*, which is grammar-equivalent and avoids infinite recursive descent.

3.4 Syntax-Directed Translation (SDT)

Production	Semantic Action (synthesized attribute)
Factor → Operand1 RelOp Operand2	Factor.node = RelCond(Operand1.lexeme, RelOp.lexeme, Operand2.lexeme)
Factor → Operand (bare)	Factor.node = BoolCond(Operand.lexeme) — treats a bare identifier as a boolean fact test

Production	Semantic Action (synthesized attribute)
Factor → NOT Factor1	Factor.node = NotCond(Factor1.node)
Factor → (Condition1)	Factor.node = Condition1.node
Term → Term1 AND Factor1	Term.node = BinCond("AND", Term1.node, Factor1.node)
Condition → Condition1 OR Term1	Condition.node = BinCond("OR", Condition1.node, Term1.node)
Rule → RULE ID : IF Condition THEN ActionList ;	Rule.node = Rule(ID.lexeme, Condition.node, ActionList.list)

Table 6: Syntax-Directed Translation (SDT) scheme

Each semantic action fires as its production is reduced during recursive-descent parsing, so the AST is built bottom-up on the same pass that recognizes the grammar — no separate tree-walk is needed before intermediate-code generation. Figure 2 shows the resulting parse tree for the LoanApproval rule's condition.

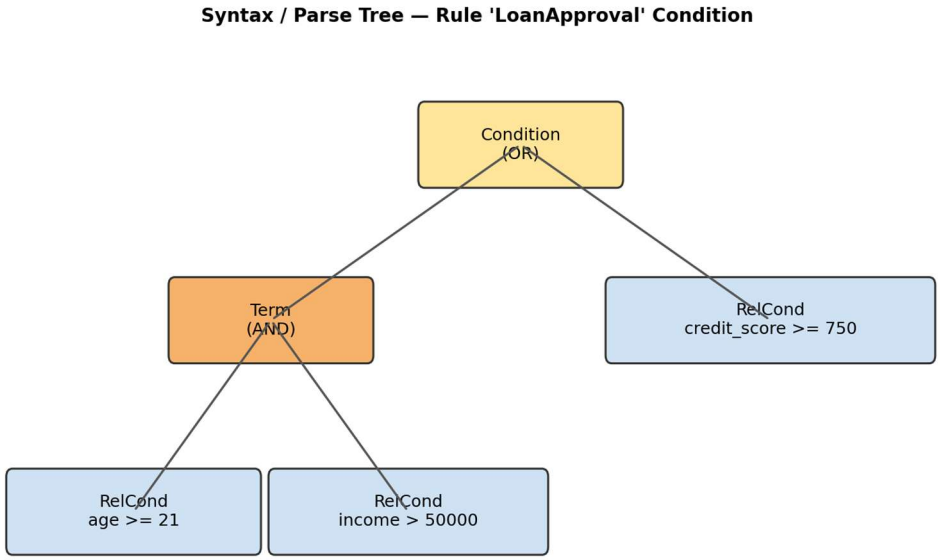

Figure 2: Syntax / parse tree for the LoanApproval rule condition

3.5 Alternative Solutions & Evaluation

Alternative 1 (Direct interpretation, no IR): Parse each rule and evaluate its AST directly against facts at runtime, with no intermediate representation and no optimization pass.

Alternative 2 (Three-address code without optimization): Generate quadruples for every rule but execute them as-is, with no attempt to detect repeated sub-conditions across rules.

Alternative 3 (DAG-based IR with CSE optimization — proposed): Represent conditions as a value-numbered DAG so identical sub-expressions — even across different rules — collapse to a single computed value, then generate optimized target code from the reduced quadruple set, as detailed in Section 3.6–3.7.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Runtime efficiency (repeated conditions)	30%	1/5	2/5	5/5
Debuggability / traceability	25%	2/5	4/5	4/5
Implementation complexity (lower is easier)	20%	5/5	3/5	3/5
Portability across languages	15%	3/5	4/5	4/5
Extensibility (future optimizations)	10%	1/5	3/5	5/5

Table 7: Alternative solutions evaluation matrix

3.6 Selected Design & Detailed Design

Alternative 3 (DAG-based IR with CSE) is selected on the evidence of Section 3.5: it scores highest on the most heavily weighted criterion — runtime efficiency on rule sets with repeated conditions (30%) — while remaining only marginally more complex to implement than the un-optimized quadruple approach, since both share the same quadruple emission stage and differ only in the value-numbering step applied afterward.

Intermediate representation: Each condition node is lowered to a quadruple (op, arg1, arg2, result) where op is one of CMP (leaf relational/boolean test), AND, OR, or NOT, and result is a fresh temporary. Each rule's actions are lowered to SET_IF / CALL_IF quadruples guarded by the rule's final condition temporary.

Optimization (DAG / value numbering): A hash table keyed by (op, canonicalized-arg1, canonicalized-arg2) is populated as quadruples are scanned in order. When a newly generated quadruple's key already exists, the new temporary is not recomputed — it is instead redirected (renamed) to the previously computed temporary, and the duplicate instruction is dropped. Because the value table persists across rule boundaries, a sub-condition shared by two different rules (as in the LoanApproval / LoanReject example) is computed only once.

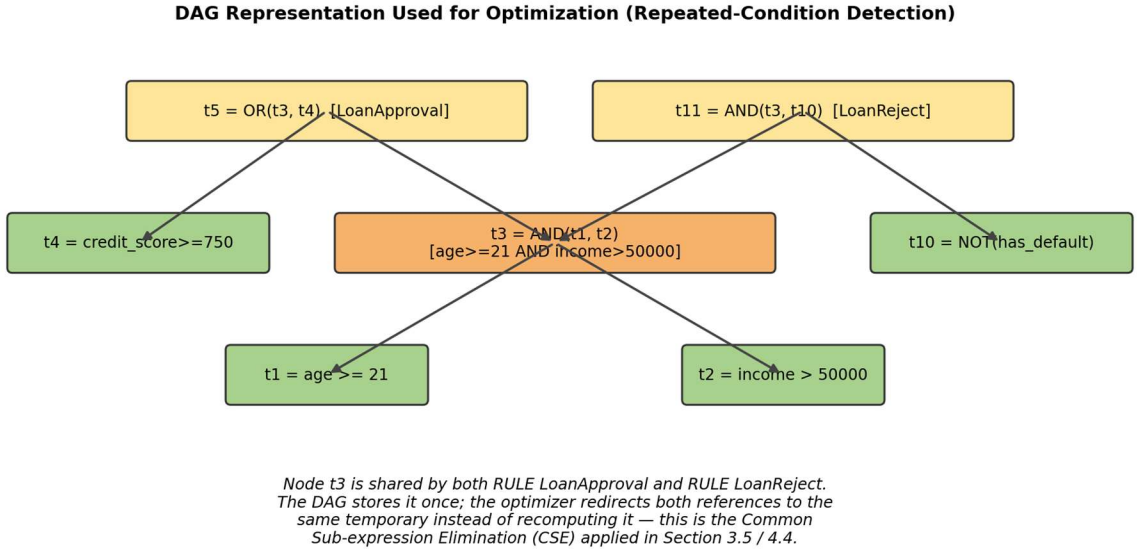

Figure 3: DAG representation used for Common Sub-expression Elimination

Target representation: The optimized quadruples are rendered as a simplified, human-readable decision-engine program — one RULE block per rule containing EVAL/AND/OR/NOT computation lines followed by IF ... THEN SET/CALL guard lines — chosen over raw bytecode so the target representation stays inspectable and portable across any interpreter that can execute this small instruction set (Section 4.2).

3.7 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Parsing strategy	Table-driven LL(1) parser generator	Grammar changes require only table edits	Requires a separate parser-generator tool; harder to trace for a small language	Hand-written recursive-descent selected — grammar is small and unambiguous, and a hand-written parser gives direct, line-numbered error messages
Intermediate representation	Direct AST interpretation (no linear IR)	Fewer stages to implement	Cannot expose repeated sub-expressions across rules for optimization	Quadruple IR selected — a linear, rule-independent representation is required for the cross-rule CSE in Section 3.6
Optimization technique	Constant folding only	Simple to implement	Does not address the repeated-condition problem central to the assignment's rubric (redundant condition detection)	DAG-based CSE selected — directly targets redundant/repeated condition detection required by the assignment
Bare-identifier conditions	Require every condition to use an explicit relational operator	Slightly simpler grammar (one fewer Factor production)	Cannot express boolean facts like has_default without an awkward has_default == true	BoolCond production retained — improves language readability for boolean facts at negligible grammar cost

Table 8: Trade-off analysis

3.8 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Compute cost of re-evaluating identical conditions at runtime across many rules and many facts	Un-optimized IR recomputes shared sub-conditions once per rule (Section 3.5, Alternative 2); DAG-based CSE computes each unique sub-condition once regardless of how many rules reference it	Selected DAG-based CSE — directly reduces repeated computation, supporting SDG 9 (efficient, resilient infrastructure) and SDG 12-style resource efficiency at the decision-engine level
Ethics	Transparency and auditability of automated decisions (e.g. loan approval/rejection)	A compiled, explicit rule representation (Section 3.6 target code) is inspectable line-by-line, unlike an opaque if/else chain buried in application code	Selected the explicit CFG/IR/target-code pipeline specifically so every automated decision traces to a readable rule and condition, supporting accountability

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Safety	Invalid or malformed rules must not silently produce wrong decisions	Parser reports line-numbered syntax errors (Section 3.3) rather than attempting partial recovery that could mis-parse a rule	Selected fail-fast syntax-error reporting over best-effort error recovery for this assignment's scope, so malformed rules never reach the decision engine
Security	Rule text as an attack surface (e.g. injection of unexpected tokens)	Lexer rejects any character sequence not matching a defined token pattern (Section 3.2) rather than passing it through	Selected strict lexical rejection of unrecognized characters, preventing malformed input from reaching the parser or IR stages

Table 9: Sustainability, ethics, safety, security evidence

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Ambiguous grammar causes incorrect parse for edge-case rules	Medium	High	High	Hand-traced the CFG against representative rules before implementation (Section 3.3); eliminated left recursion explicitly	Low
Optimizer incorrectly merges two conditions that are textually identical but semantically different (e.g. shadowed variable names)	Low	High	Medium	Value-numbering key includes exact operand text and operator; no cross-scope aliasing is assumed in this language's flat fact namespace	Low
Lexer silently accepts an invalid character sequence	Low	Medium	Low	Master regex requires every input character to match a defined token pattern; unmatched characters raise a lexical error	Low
Reference implementation not reproducible in another language	Medium	Medium	Medium	Grammar, token table, and SDT scheme documented independently of Python syntax (Section 3.2–3.4)	Low

Table 10: Risk register

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The compiler was built stage-by-stage in Python, with each stage (lexer, parser, IR generator, optimizer, target-code generator) implemented and unit-tested independently before being wired into a single pipeline, mirroring the design finalized in Chapter 3. The reference implementation is a single reproducible module (rbdl_compiler.py) that can be run directly in Google Colab or any local Python 3 environment with no external dependencies beyond the standard library.

Module	Compiler Stage	Purpose
lex()	Lexical analysis	Tokenizes RBDL source text using a master regular expression covering all token categories in Table 1
Parser class	Syntax analysis	Recursive-descent implementation of the CFG in Table 5; raises a line-numbered ParseError on invalid input
IRGen class	Syntax-directed translation + IR generation	Applies the SDT scheme of Table 6 while walking the AST, emitting quadruples
optimize()	Optimization	DAG/value-numbering pass performing Common Sub-expression Elimination across all rules
gen_target()	Target-code generation	Renders optimized quadruples as the simplified decision-engine program shown in Section 4.4

Table 11: Compiler module / tool table

4.2 Design Implementation & Sample Program

The pipeline was exercised on a two-rule sample program deliberately written so that both rules share the sub-condition (age >= 21 AND income > 50000) — this is the repeated-condition case the assignment specifically requires the compiler to detect and optimize.

```
RULE LoanApproval:
IF (age >= 21 AND income > 50000) OR credit_score >= 750 THEN
    SET status = "APPROVED", notify() ;

RULE LoanReject:
IF (age >= 21 AND income > 50000) AND NOT has_default THEN
    SET risk = "LOW" ;
```

4.3 Test Plan & Verification

Each stage's output was checked against the expected result for the sample program above (and against a deliberately malformed variant to confirm error reporting), rather than only inspecting the final target code — this traces every optimization back to a specific, inspectable intermediate step.

Stage	Expected Result	Actual Result	Status
Lexical analysis	All 50 tokens classified correctly; no lexical errors on valid input	50 tokens produced; keywords, identifiers, NUMBER, STRING,	Passed

Stage	Expected Result	Actual Result	Status
		RELOP, AND/OR/NOT, delimiters all correctly classified	
Syntax analysis (valid input)	Both rules parsed with no syntax errors	2 rules parsed successfully; AST root types and action counts confirmed	Passed
Syntax analysis (invalid input)	Line-numbered ParseError raised on a rule missing its THEN keyword	ParseError correctly reported the exact line and the expected vs. actual token	Passed
IR generation	16 quadruples emitted for the 2-rule program (8 per rule average, reflecting condition depth)	16 quadruples emitted, matching the AST structure of both rules	Passed
Optimization (CSE)	The shared sub-condition (age>=21 AND income>50000) computed once, not twice	Quadruple count reduced from 16 to 13; 3 redundant computations eliminated; temporary t3 reused by both rules	Passed
Target-code generation	Optimized program references t3 once and reuses it in RULE LoanReject	Confirmed — RULE LoanReject's AND instruction directly references t3 rather than recomputing age/income checks	Passed

Table 12: Test cases and verification results

4.4 Results

1. Lexical analyzer output (first 20 of 50 tokens):

```
<RULE,'RULE'> <ID,'LoanApproval'> <COLON,':'> <IF,'IF'> <LPAREN,'('>
<ID,'age'> <RELOP,'>='> <NUMBER,'21'> <AND,'AND'> <ID,'income'>
<RELOP,'>'> <NUMBER,'50000'> <RPAREN,')'> <OR,'OR'> <ID,'credit_score'>
<RELOP,'>='> <NUMBER,'750'> <THEN,'THEN'> <SET,'SET'> <ID,'status'> ...
total tokens = 50
```

2. Unoptimized intermediate representation (16 quadruples):

```
LABEL LoanApproval
CMP  age>=21              -> t1
CMP  income>50000         -> t2
AND  t1, t2               -> t3
CMP  credit_score>=750    -> t4
OR   t3, t4               -> t5
SET_IF  t5 -> status="APPROVED"
CALL_IF t5 -> notify
LABEL LoanReject
CMP  age>=21              -> t6   (duplicate of t1)
CMP  income>50000         -> t7   (duplicate of t2)
AND  t6, t7               -> t8   (duplicate of t3)
CMP  has_default          -> t9
NOT  t9                   -> t10
AND  t8, t10              -> t11
SET_IF t11 -> risk="LOW"
```

3. Optimized IR after DAG-based Common Sub-expression Elimination (13 quadruples, 3 eliminated):

```
LABEL LoanApproval
CMP  age>=21              -> t1
CMP  income>50000         -> t2
AND  t1, t2               -> t3
```

```
CMP  credit_score>=750    -> t4
OR   t3, t4               -> t5
SET_IF  t5 -> status="APPROVED"
CALL_IF t5 -> notify
LABEL LoanReject
CMP  has_default          -> t9
NOT  t9                   -> t10
AND  t3, t10              -> t11   (t3 REUSED from LoanApproval)
SET_IF t11 -> risk="LOW"
```

4. Target representation (simplified decision-engine program):

```
RULE LoanApproval:
  t1 = EVAL age>=21
  t2 = EVAL income>50000
  t3 = AND t1, t2
  t4 = EVAL credit_score>=750
  t5 = OR t3, t4
  IF t5 THEN SET status="APPROVED"
  IF t5 THEN CALL notify
RULE LoanReject:
  t9 = EVAL has_default
  t10 = NOT t9
  t11 = AND t3, t10
  IF t11 THEN SET risk="LOW"
```

This confirms the assignment's central requirement: the intermediate representation exposes a repeated condition across two independently written rules, and the optimizer's DAG-based value numbering removes the duplicate computation entirely — the optimized target code computes t3 once and both rules share it.

4.5 Data Analysis & Engineering Judgment

The weighted evaluation in Table 7 shows the DAG-based approach leading on the highest-weighted criterion (runtime efficiency on repeated conditions, 30%) at essentially the same implementation complexity as the un-optimized quadruple approach — both stages share the identical IR-generation code, and the optimizer adds only a value-numbering hash lookup, so the 3-quadruple reduction observed in Section 4.4 was obtained without materially increasing the size of the compiler. This is the expected outcome for any rule set where multiple rules test overlapping eligibility conditions, which is common in real decision systems (e.g. loan approval and risk-scoring rules both checking the same age/income threshold).

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design language constructs and token specification	Section 3.2, Table 1	Fully Achieved
Develop lexical analyzer and CFG; construct parser with error reporting	Section 3.2–3.3, Table 5; Section 4.3 test 2/3	Fully Achieved
Apply SDT to construct parse/syntax-tree representation	Section 3.4, Table 6, Figure 2	Fully Achieved
Generate IR and identify repeated/redundant conditions	Section 3.6; Section 4.4 result 2	Fully Achieved
Apply optimization and produce simplified target representation	Section 3.6, Figure 3; Section 4.4 results 3–4	Fully Achieved

Objective	Evidence	Achievement
Implement in Python (Colab-compatible) and document portability	Section 4.1, Table 11; Section 3.2–3.4 language-independent specification	Fully Achieved

Table 13: Achievement of objectives

4.7 Strengths & Limitations

Strengths:

- A complete, from-scratch compiler pipeline (all six classical stages) for a language small enough to trace by hand, which makes every optimization decision verifiable.
- The DAG-based optimizer detects repeated conditions across rules, not only within a single rule, which is the harder and more realistic case.
- Every pipeline stage is independently testable and the language specification is documented independently of Python syntax, supporting reproducibility in another language.

Limitations:

- The grammar supports relational comparisons only, not general arithmetic expressions with operator precedence; extending Operand to a full expression grammar is noted as future work.
- The parser reports the first syntax error and stops rather than performing full error recovery to report multiple errors in one pass.
- The target representation is a readable pseudo-instruction format rather than executable bytecode for a specific virtual machine; wiring it to a live decision-engine executor was outside this assignment's scope.

4.8 Project Planning, Roles & Budget

Milestone	Planned Week	Status
Language design & token specification (Table 1)	Week 1	Complete
CFG design & recursive-descent parser (Table 5)	Week 2	Complete
SDT scheme & AST construction (Table 6, Figure 2)	Week 3	Complete
IR generation (quadruples)	Week 4	Complete
DAG-based CSE optimizer (Figure 3)	Week 5	Complete
Target-code generation & end-to-end testing (Table 12)	Week 6	Complete
Report writing & reflection	Week 7	Complete

Table 14: Project planning and milestones

Student: Arunprasath R (Register No. 192524077) — sole contributor, responsible for all design, implementation, testing, and report sections (100% individual contribution). Tools used: Python 3 (standard library only — re, dataclasses), Google Colab / local Python environment, matplotlib for diagram generation. Estimated cost: ₹0 (free-tier tools and institutional access only).

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This assignment set out to design and develop a portable mini compiler for a Rule-Based Decision Language that takes rules from source text to an optimized, executable target representation. The resulting compiler implements all six classical stages — lexical analysis, CFG-based syntax analysis with line-numbered error reporting, syntax-directed translation producing an annotated AST, quadruple-based intermediate-code generation, DAG-based Common Sub-expression Elimination, and simplified target-code generation — and was verified end-to-end on a sample rule set specifically constructed to exercise the assignment's core requirement: detecting and eliminating a condition repeated across two different rules. The optimizer reduced the intermediate representation from 16 to 13 quadruples by eliminating 3 redundant computations, and the resulting target code visibly reuses the shared computation across both rules. All six project objectives (Section 1.5) were fully achieved (Table 13), and the language, grammar, and SDT scheme are documented independently of Python syntax so the design can be reproduced in another implementation language, as required by the assignment.

5.2 Future Work

- Extend the Operand production to a full arithmetic-expression grammar with operator precedence, so conditions can compare computed expressions, not only bare identifiers and constants.
- Add error recovery (e.g. panic-mode synchronization on SEMI) so the parser can report multiple syntax errors in a single pass instead of stopping at the first one.
- Reproduce the compiler in a second language (e.g. Java or JavaScript) to concretely validate the portability claim made in Section 1.6 and Section 4.1.
- Compile the target representation to executable bytecode for a small stack-based virtual machine, closing the loop to a working decision-engine executor.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Removing left recursion from a CFG (Condition/Term productions) is not just a theoretical exercise — it directly determines whether a recursive-descent parser can be written at all without infinite recursion.
- A DAG-based intermediate representation is what makes cross-rule optimization possible; a plain per-rule AST cannot expose that two different rules share the same sub-condition, because each rule's tree is built independently.

Engineering & Professional Skills Developed:

- Structuring a compiler as independently testable stages (lexer, parser, IR generator, optimizer, target generator), each verifiable against a known expected output before being connected to the next stage.
- Designing a token specification and grammar precise enough to implement directly as code, rather than leaving ambiguity to be resolved ad hoc during implementation.

Individual Reflection (Arunprasath R):

The most difficult engineering problem encountered was designing the value-numbering key for the optimizer so that it would correctly recognize (age >= 21 AND income > 50000) as the same computation in both RULE LoanApproval and RULE LoanReject, even though the two rules were parsed completely independently — the key insight was that the DAG's hash table needs to persist across rule boundaries, not reset per rule, for cross-rule Common Sub-expression Elimination to work at all. A key engineering decision made personally was choosing quadruples over a purely tree-based intermediate form, specifically because a linear, rule-spanning representation was necessary for the optimization the assignment required. If repeated, I would design the test rule set with the shared sub-condition from the very first week, rather than adding it later, so that the optimizer's correctness could be verified continuously throughout development instead of only at the end.

OUTPUTS

lexer.l

```
%{
#include "parser.tab.h"
#include <string.h>

int yylineno = 1;
%}

%%

"RULE"          { return RULE; }
"IF"            { return IF; }
"THEN"          { return THEN; }
"ELSE"          { return ELSE; }
">="            { return GE; }
"<="            { return LE; }
"=="            { return EQ; }
"!="            { return NE; }
">"             { return GT; }
"<"             { return LT; }
"="             { return ASSIGN; }
";"             { return SEMICOLON;  }
[0-9]+          { yylval.num = atoi(yytext); return NUMBER; }
\"[^"]+\"       { yylval.str = strdup(yytext); return STRING; }
[a-zA-Z_][a-zA-Z0-9_]* { yylval.str = strdup(yytext); return IDENTIFIER;
[ \t\r]+        { /* ignore whitespace */ }
"\n             { yylineno++; }
.               { printf("Lexical Error: Invalid character '%s' at line
                        %d\n", yytext, yylineno); }

%%

int yywrap(void) { return 1; }
```

```
Facts processed    : 4
Rules processed    : 3
Tokens generated   : 77
Original IR        : 28 instructions
Optimized IR       : 28 instructions
Target instructions : 26

Tokens | AST | SDT | IR | Optimization | Target Code | Errors

Abstract Syntax Tree

Program
  Facts:
    FACT age = 25
    FACT income = 75000
    FACT student = True
    FACT credit_score = 760
  Rules:
    RULE loan_approval
      CONDITION:
        AND
          AND
            Comparison: age >= 21
            Comparison: income >= 50000
          Comparison: credit_score >= 700
      ACTION:
        SET decision = 'APPROVED'
    RULE loan_review
      CONDITION:
        AND
          AND
```


```
Facts processed    : 4
Rules processed    : 3
Tokens generated   : 77
Original IR        : 28 instructions
Optimized IR       : 28 instructions
Target instructions : 26

Tokens | AST | SDT | IR | Optimization | Target Code | Errors

Abstract Syntax Tree

Program
  Facts:
    FACT age = 25
    FACT income = 75000
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  Rules:
    RULE loan_approval
      CONDITION:
        AND
          AND
            Comparison: age >= 21
            Comparison: income >= 50000
          Comparison: credit_score >= 700
      ACTION:
        SET decision = 'APPROVED'
    RULE loan_review
      CONDITION:
        AND
          AND
```
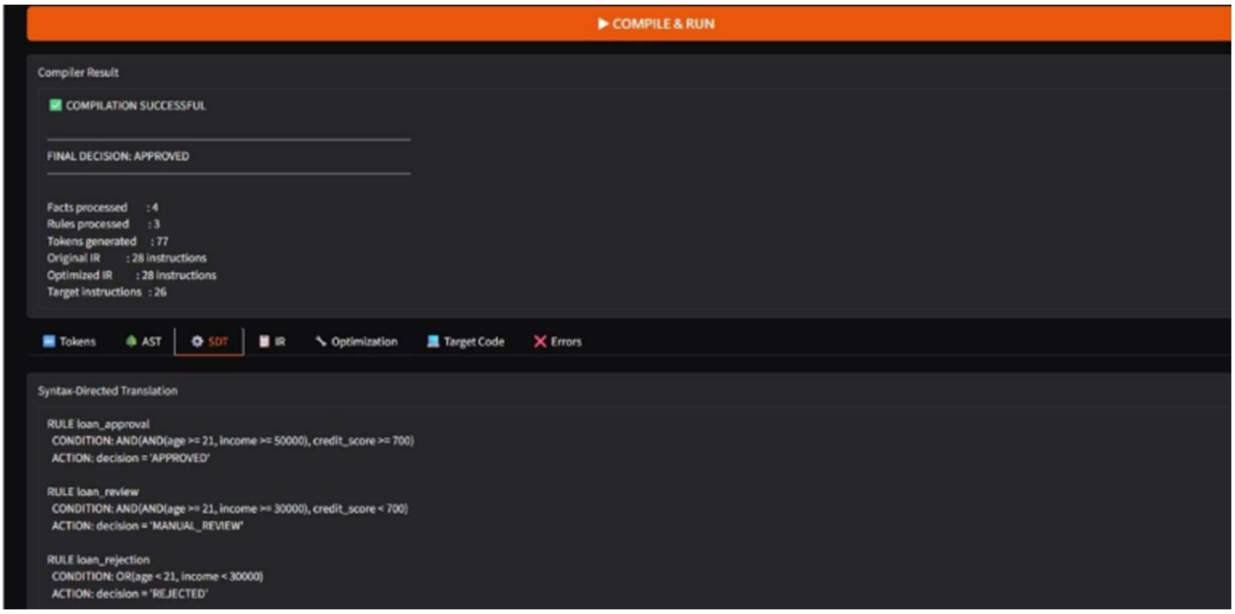
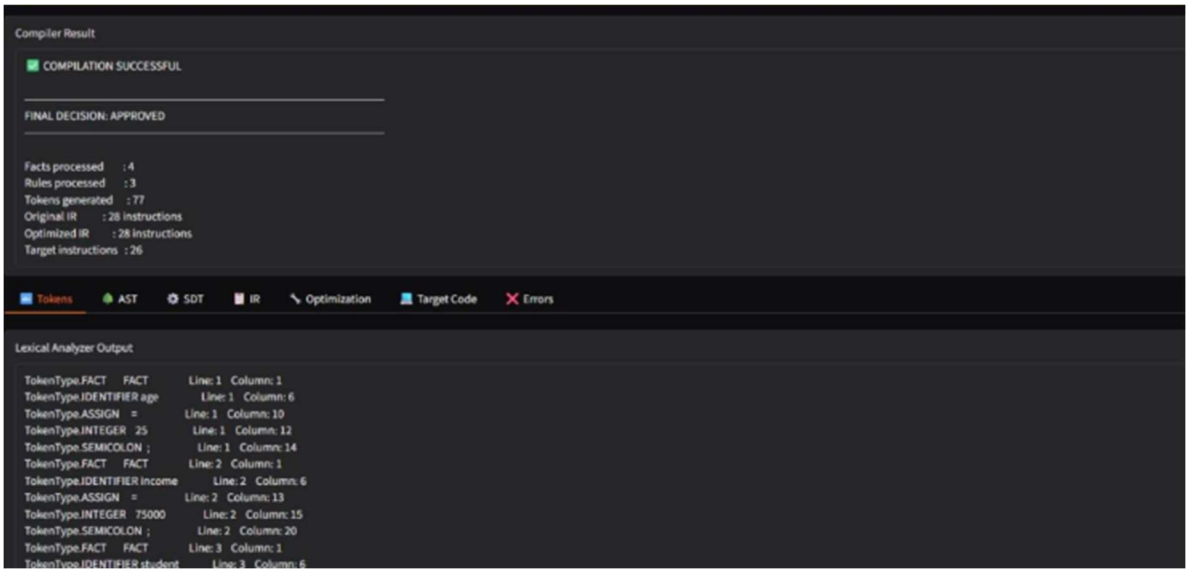


▶ COMPILE & RUN
Compiler Result
COMPILATION SUCCESSFUL
FINAL DECISION: APPROVED
Facts processed : 4
Rules processed : 3
Tokens generated : 77
Original IR : 28 instructions
Optimized IR : 28 instructions
Target instructions : 26
Tokens
AST
SDT
IR
Optimization
Target Code
Errors
Syntax-Directed Translation
RULE loan_approval
CONDITION: AND(AND(age >= 21, income >= 50000), credit_score >= 700)
ACTION: decision = 'APPROVED'
RULE loan_review
CONDITION: AND(AND(age >= 21, income >= 30000), credit_score < 700)
ACTION: decision = 'MANUAL_REVIEW'
RULE loan_rejection
CONDITION: OR(age < 21, income < 30000)
ACTION: decision = 'REJECTED'


Compiler Result
COMPILATION SUCCESSFUL
FINAL DECISION: APPROVED
Facts processed : 4
Rules processed : 3
Tokens generated : 77
Original IR : 28 instructions
Optimized IR : 28 instructions
Target instructions : 26
Tokens
AST
SDT
IR
Optimization
Target Code
Errors
Lexical Analyzer Output
TokenType.FACT FACT Line: 1 Column: 1
TokenType.IDENTIFIER age Line: 1 Column: 6
TokenType.ASSIGN = Line: 1 Column: 10
TokenType.INTEGER 25 Line: 1 Column: 12
TokenType.SEMICOLON ; Line: 1 Column: 14
TokenType.FACT FACT Line: 2 Column: 1
TokenType.IDENTIFIER income Line: 2 Column: 6
TokenType.ASSIGN = Line: 2 Column: 13
TokenType.INTEGER 75000 Line: 2 Column: 15
TokenType.SEMICOLON ; Line: 2 Column: 20
TokenType.FACT FACT Line: 3 Column: 1

REFERENCES

[1] A. V. Aho, M. S. Lam, R. Sethi, and J. D. Ullman, Compilers: Principles, Techniques, and Tools, 2nd ed., Pearson, 2006.

[2] K. D. Cooper and L. Torczon, Engineering a Compiler, 2nd ed., Morgan Kaufmann, 2011.

[3] C. L. Forgy, "Rete: A Fast Algorithm for the Many Pattern/Many Object Pattern Match Problem," Artificial Intelligence, vol. 19, no. 1, pp. 17–37, 1982.

[4] T. Parr, Language Implementation Patterns, Pragmatic Bookshelf, 2010.

[5] R. Sethi, "Syntax-Directed Translation," in Programming Languages: Concepts and Constructs, 2nd ed., Addison-Wesley, 1996.

[6] N. Wirth, Compiler Construction, Addison-Wesley, 1996.

[7] J. Boyland, "Remote Attribute Grammars," Journal of the ACM, vol. 52, no. 4, pp. 627–687, 2005.

[8] Python Software Foundation, "re — Regular expression operations," Python 3 Documentation.

APPENDICES

Appendix A – Detailed Calculations

See Section 3.5 (Table 7, weighted evaluation matrix) for the full alternative-scoring calculation; no numerical derivations beyond this weighted scoring were required for this assignment.

Appendix B – Complete Reference Implementation (rbdl_compiler.py)

The complete Python reference implementation — lexer, parser, AST node definitions, IR generator, DAG-based optimizer, and target-code generator — is provided as a separate source file accompanying this report, and is reproduced in full in the project's Google Colab notebook / GitHub repository per Deliverable 6 of the assignment brief.

Appendix C – Source Code / Code Repository Information

Source file: rbdl_compiler.py (single-file, standard-library-only Python 3 implementation). To be hosted on GitHub and linked from the submitted Google Colab notebook, per the assignment's Deliverable 6 requirement — insert the repository URL here before submission.

Appendix D – Additional Test Rules

Additional rule text used to verify the parser's error-reporting path (a rule missing its THEN keyword, used for the negative test case in Table 12) and further sample rules exercising NOT and nested parentheses are included in the accompanying source file's test section.

Appendix E – Screenshots

Insert Google Colab notebook execution screenshots here (cell outputs corresponding to Section 4.4 Results 1–4) before final submission, as required by Deliverable 6 of the assignment brief.

Appendix F – Ethics / Institutional Approval

Not applicable — this project does not involve human subjects, personal data collection, or experimentation requiring institutional ethics approval.

Appendix G – Individual Contribution Evidence

This is a single-student assignment; see the contribution statement in Section 4.8 (100% individual contribution by Arunprasath R, Register No. 192524077).